Image Processing Practice Sessions

Lab Report

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1 Part 1 : Basic operations on images

1.1 Image histogram

1 – Display the histogram of each image

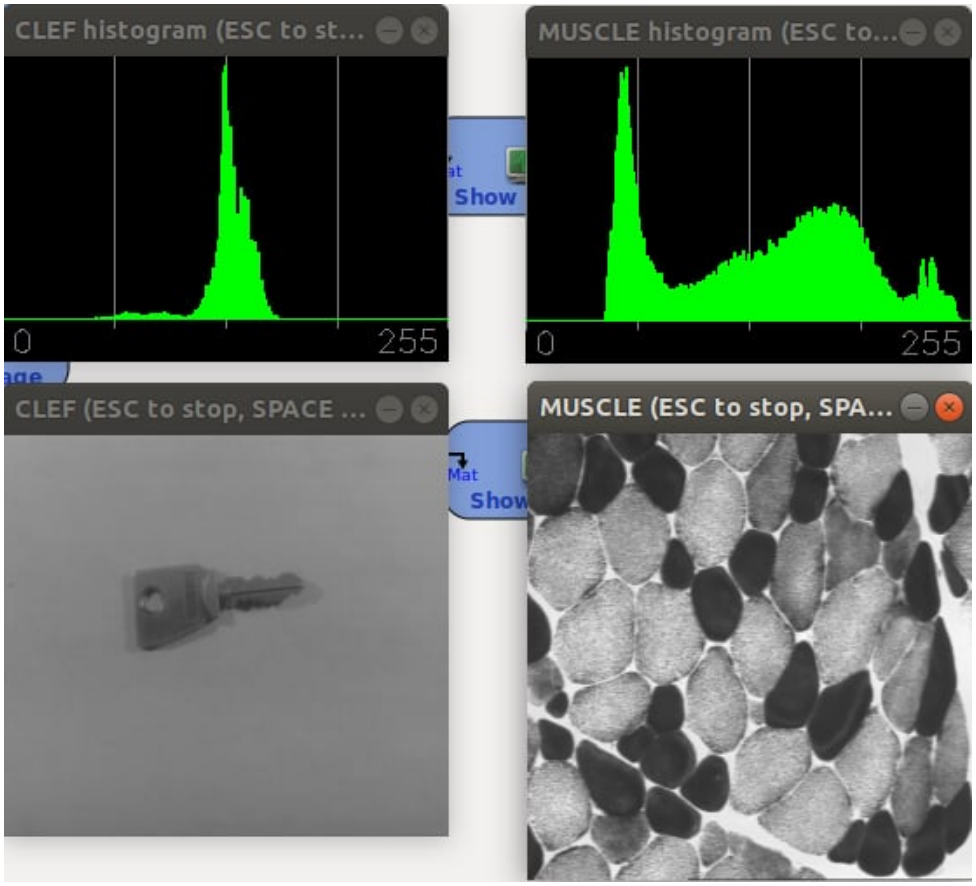


Figure 1: Histograms of `muscle.bmp` and `clef.bmp`.

2 – Explain the shape of each histogram

CLEF.
The key histogram has a unimodal distribution:

- The very tall peak represents the pixels belonging to the background, which has a white gray colour. Since the background covers a large portion of the image, it dominates the pixel distribution.
- The smaller region on the middle left represents the darker pixels of the key itself.

MUSCLE.
The muscle histogram has a bimodal distribution instead; the two peaks are separated and of different heights:

- The peak on the left (closer to 0 / black) represents the pixels of the darker muscle fibres. As the height of the peak is very distinctive, we can deduce that the darker muscle fibres share the same shade of colour more often compared to the lighter counterparts.

- The peak on the right (closer to 255 / white) represents the pixels of the lighter muscle fibres as well as the surrounding space. This peak is rounder and flatter, indicating that the shades of the lighter muscle fibres are more distributed rather than focused.
- The roughly similar heights of the peaks also suggest that the dark and light fibres occupy similar proportions of the image area.

3 – Using only `muscle.bmp` histogram

Which fibres are more dense?
The fibre type associated with the higher peak is the more dense one, because a higher peak means more pixels at those intensities. In this histogram, the dark-fibre peak is higher, so dark fibres are slightly more abundant in the image.

Which fibres have a more homogeneous colour?
The peak of the darker muscle fibres is narrower and more peaked, indicating that the pixels share similar brightness. Therefore, dark fibres have a more homogeneous colour.

1.2 Arithmetic operations

1–3: Histograms of the two images and of their sum

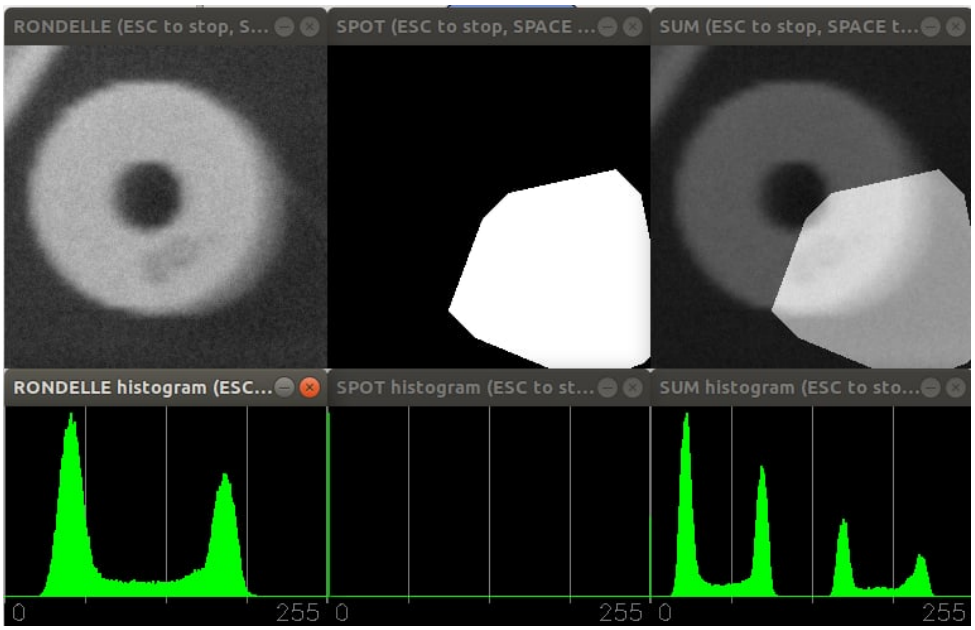


Figure 2: Histograms of `rondelle.bmp`, `spot.bmp` and their sum.

Why do we use a "multiply by scalar" operation before adding the images?
By multiplying each image by 0.5, it reduces their value range from $[0, 255]$ to $[0, 127.5]$. When the two images are added, the result stays within $[0, 255]$. This avoids saturation of values above 255, which would otherwise all be truncated to 255 and destroy information in bright regions.

Explain the shape of the histograms.

RONDELLE.

The `rondelle.bmp` histogram has a bimodal distribution:

- The higher left peak represents the larger number of pixels in the darker background and the dark circle in the centre of the image.
- The shorter right peak represents the pixels of the bright gray ring.

SPOT.

The `spot.bmp` histogram also has a bimodal distribution. However, the two peaks are separated and extremely narrow with almost zero values between them:

- The seemingly one-line peak on the left represents the vast majority of the image area, which is the black background of the mask.
- The seemingly one-line peak on the right represents the white spot.

This signifies that the colours of the background and of the spot are almost pure.

SUM (RONDELLE + SPOT).

After scaling each image by 0.5 and adding them, the resulting histogram becomes multimodal:

- Dark background regions of `rondelle` combined with dark regions of `spot` produce a darker lobe on the left.
- Bright regions of the spot overlapping the ring produce brighter values, forming peaks on the right.

These different combinations of dark/bright areas in the two images explain the multiple peaks.

4 – Remove the "multiply by scalar" operations

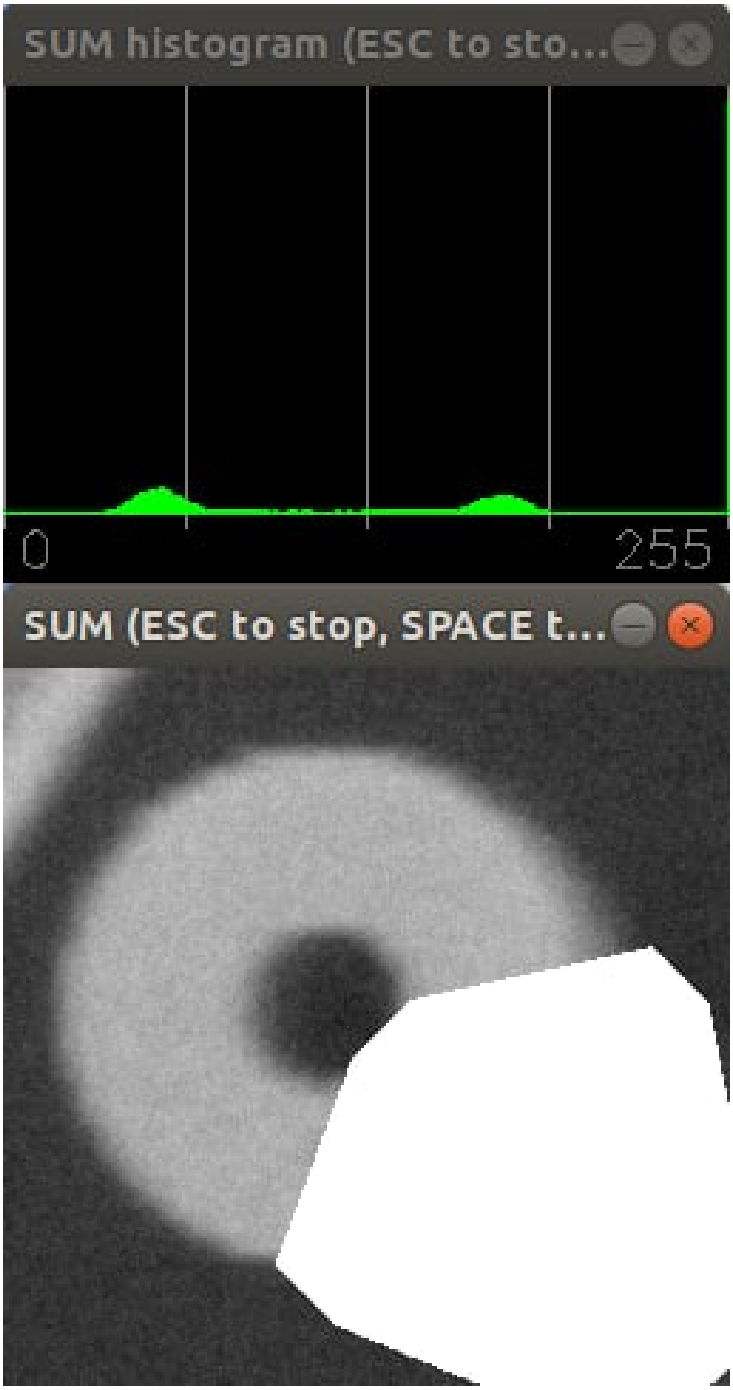


Figure 3: Histogram of the sum without scaling.

The histogram has a bimodal shape with a distinct line on the right edge of the distribution. Without the scalar, some sums exceed 255. Because the image is stored with 8-bit pixels, all values above 255 are clipped to 255. This creates a very tall bar at the rightmost intensity (255) in the histogram, corresponding to all saturated pixels. The rest of the distribution looks similar to the original `rondelle` histogram but with less contrast, because many bright combinations have been flattened to the same maximum value.

5 – Can we predict the histogram of the sum using only the two histograms?

No, knowing only the histograms of two images is not enough to predict the histogram of their sum:

- A histogram shows pixel counts per intensity, but not which pixels correspond (which pixel from the first image goes with which pixel from the second image).
- The resulting histogram depends on how the pixels are paired during addition.
- Different pixel arrangements with the same histograms can produce different sum histograms.

1.3 Negative image

1–3: Construction and histograms

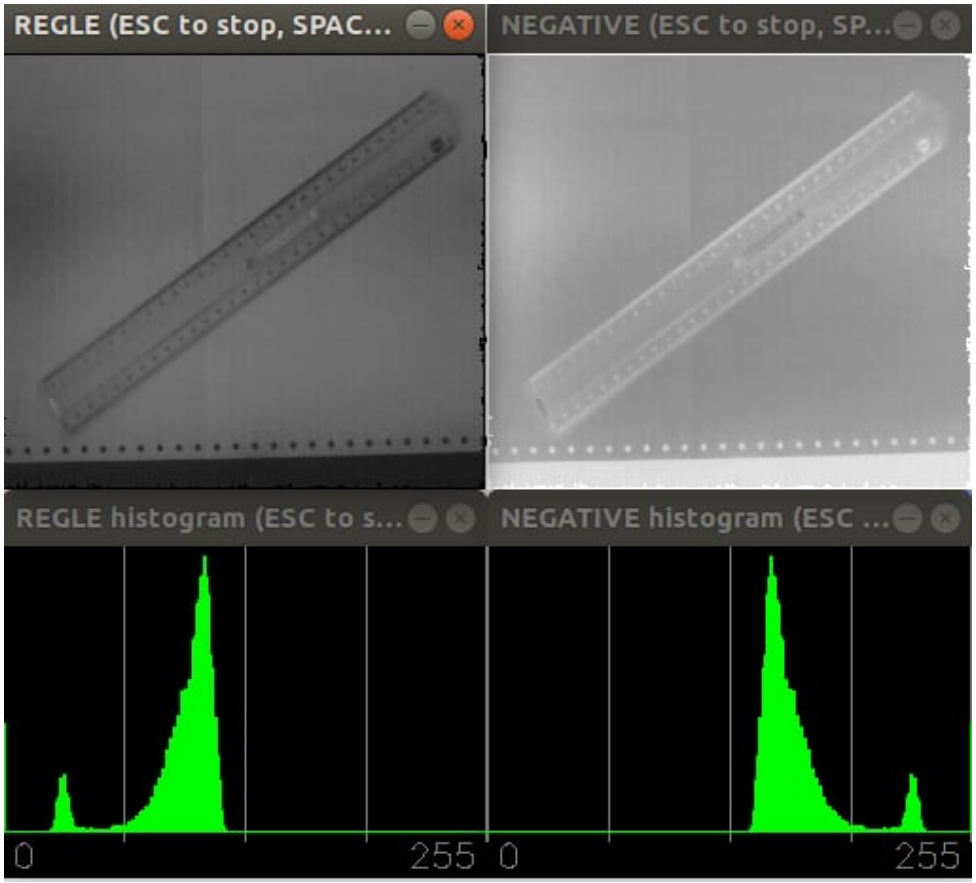


Figure 4: Histogram of `regle.bmp` and of its negative.

We create a new image with all pixels set to 255, then subtract the original image `regle.bmp` from this image. The negative image is computed as

$$g(x, y) = 255 - f(x, y).$$

Explain the shape of the histogram of the resulting image.

Each gray level i in the original image is mapped to $255 - i$. As a result, the histogram of the negative image is a horizontal mirror of the original histogram: the frequency of each gray level is preserved, but intensities are reversed (dark becomes bright and bright becomes dark).

1.4 Quantification

Exercise (a)

Figure 5: Image `bureau.bmp` quantised on 8, 5, 3 and 1 bits.

1 – Quantify the image on 5, 3 and 1 bits.

2 – Conclusion. By reducing the number of bits, we reduce the number of available gray levels. As a result, the image shows fewer distinct shades of gray and visible contouring appears, especially at 1 bit where the image becomes purely black and white.

3 – Histogram of the image quantified on 3 bits.

Figure 6: Histogram of `bureau.bmp` quantified on 3 bits.

The number of possible shades in an image is 2^n, where n is the number of bits:

- for an 8-bit image: $2^8 = 256$ grey levels,
- for a 3-bit image: $2^3 = 8$ grey levels only.

The histogram therefore contains exactly eight bars, each located at one of these discrete gray levels. These levels are evenly spaced along the horizontal axis, but their heights depend on the image content.

Exercise (b): Least significant and most significant bit planes

For a typical 8-bit grayscale image, a pixel value P is represented by eight bits:

$$b_7, b_6, b_5, b_4, b_3, b_2, b_1, b_0.$$

Most significant bit plane (MSB).

This is the plane corresponding to the 7th bit b_7, which is the most significant bit. Its weight is $2^7 = 128$.

The MSB plane can be obtained using a bitwise AND operation with the mask `10000000` (128 in decimal), which keeps only the b_7 bit for each pixel, effectively quantising the image to 1 bit.

Least significant bit plane (LSB).

This is the plane corresponding to the 0th bit b_0, which is the least significant bit. Its weight is $2^0 = 1$.

The LSB plane can be obtained using a bitwise AND operation with the mask `00000001` (1 in decimal). It mostly contains fine noise and small variations that have little visual impact on the original image.

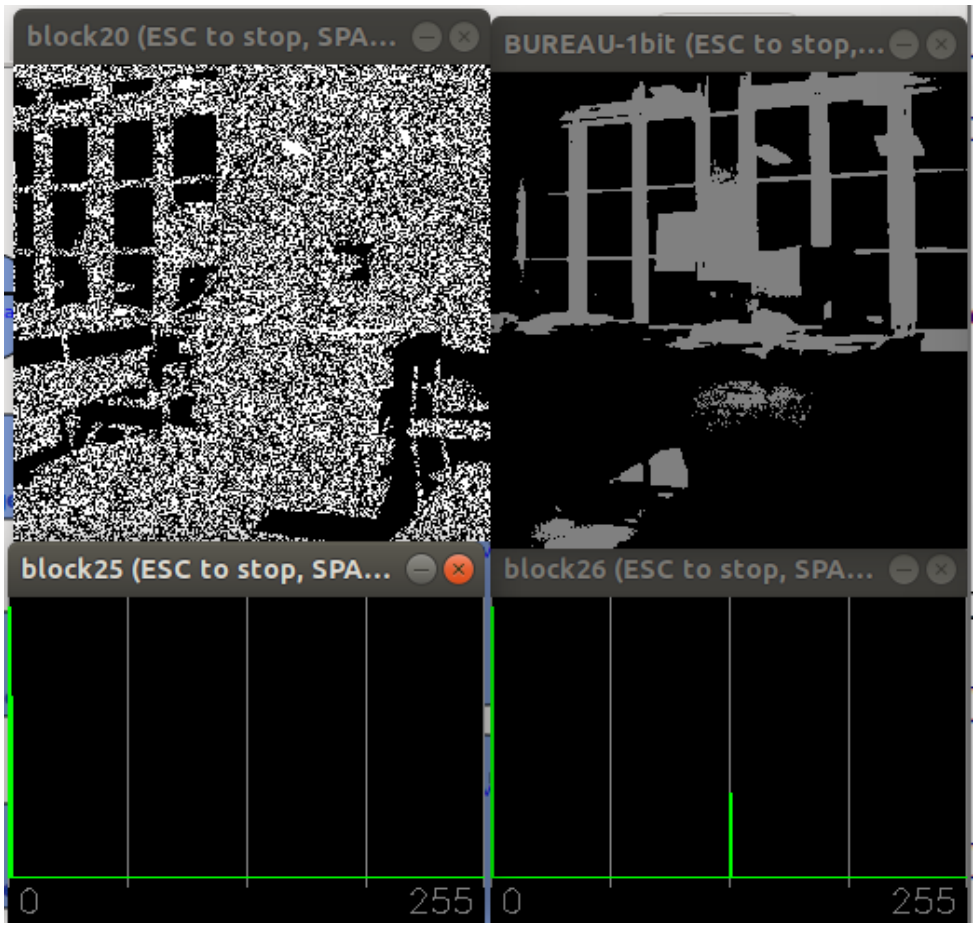


Figure 7: Most significant bit (MSB) plane and least significant bit (LSB) plane.

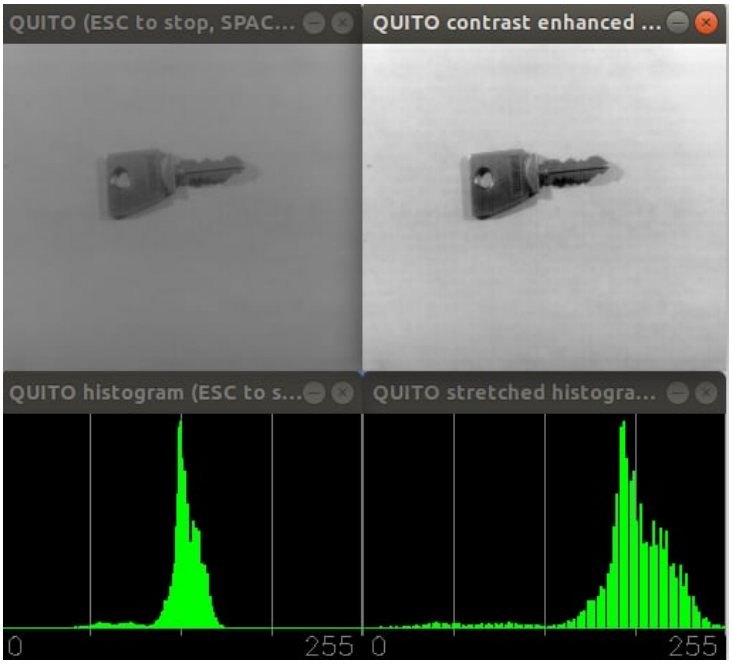
2 Part 2: Histogram Transformations

2.1 Image Enhancement with Histogram Modification

Exercise (a)

1–2. Histogram stretching on `clef.bmp`

The histogram stretching operation expands the dynamic range of the input image by mapping the minimum and maximum intensities to 0 and 255 respectively.



3. Formula relating input and output gray levels

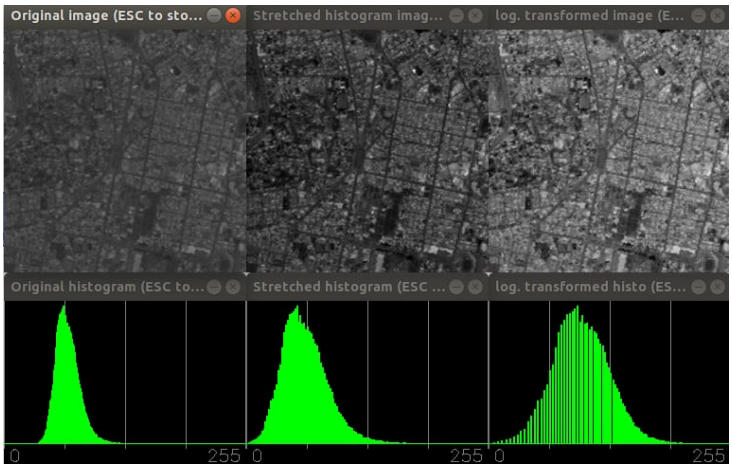
If a is the input intensity, $a_{\min}$ and $a_{\max}$ the minimum and maximum gray levels found in the image, the transformation applied is:

$$b = \frac{a - a_{\min}}{a_{\max} - a_{\min}} \times 255.$$

This linear scaling redistributes gray levels over the full available range, increasing the perceived contrast.

Exercise (b)

1. Histogram stretching followed by logarithmic transform on `quito.bmp`



The histogram stretching expands the intensity range, improving overall contrast. After that, a logarithmic transform is applied:

$$s = c \cdot \log(1 + r)$$

where r is the input pixel value normalized to $[0, 1]$, and

$$c = \frac{255}{\log(256)}$$

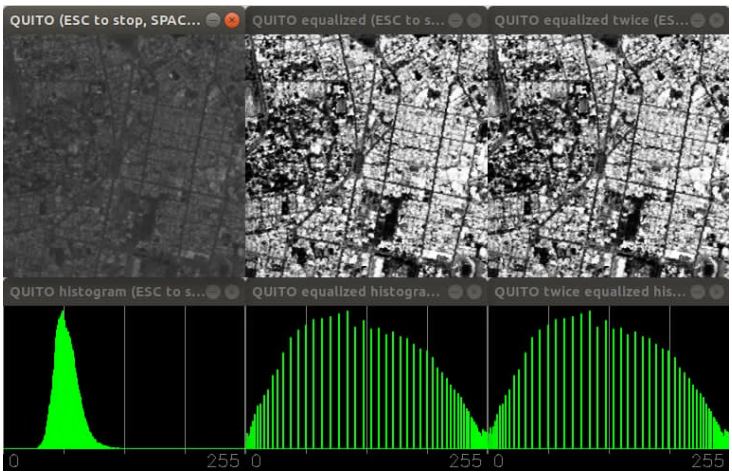
so that the output spans the full dynamic range.

The log transform expands dark intensities while compressing bright ones. As a result, the histogram is shifted toward mid-intensity values, making dark regions more visible and reducing contrast in brighter areas.

2.2 Histogram Equalization

Exercise (a)

1-3. Original histogram and after first and second histogram equalization



Histogram equalization uses the cumulative distribution function (CDF) of the image histogram to redistribute pixel intensities so that the resulting histogram becomes approximately uniform. The transformation is:

$$s_k = (L-1)\sum_{j=0}^{k} p(r_j)$$

where $p(r_j)$ is the probability of gray level r_j and $L = 256$.

The equalized histogram spreads the intensities more evenly and increases global contrast.

After the first equalization, the histogram is already close to uniform; its CDF becomes nearly linear. Applying equalization again therefore produces almost no change.

4. Repeating steps 1–3 on `lena.bmp`

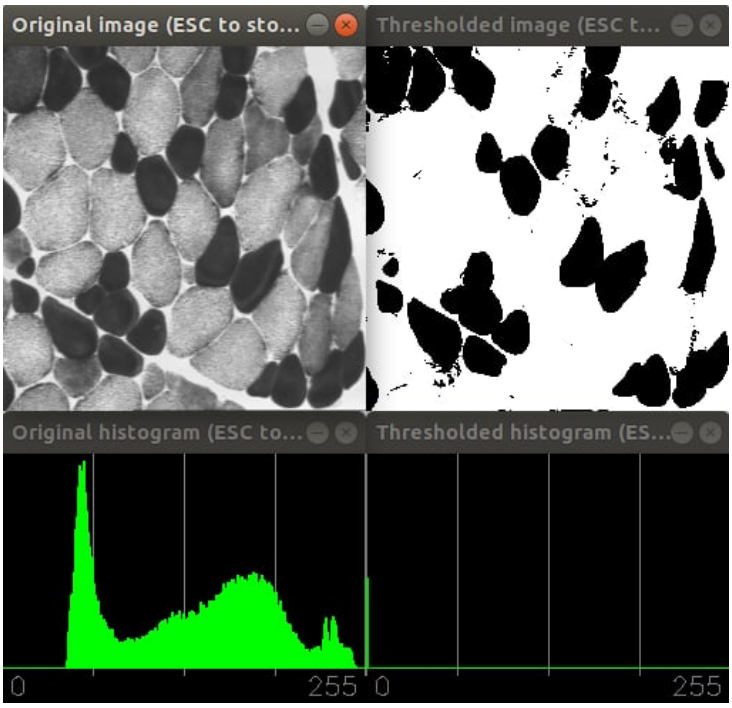
The same behavior is observed:

First equalization:

- The image:
  - Contrast becomes higher (Dark areas get darker, bright areas get brighter)
  - The features become more distinctive, sharper
- The historgram:
  - It is spread across 0-255
  - The histogram becomes more uniform while retaining partly the form of its origional shape, fewer gaps between the frequency lines

2.3 Thresholding

1. Histogram of `muscle.bmp` and threshold selection



The histogram shows two distinct groups of intensities, corresponding to dark fibers and brighter surrounding tissue. The valley between the two peaks occurs around intensity $T \approx 90$, which is an appropriate threshold to separate the dark fibers.

2. Thresholded image and histogram After applying the threshold:

$$I_{\text{out}}(x, y) = \begin{cases} 0 & \text{if } I(x, y) < T \\ 255 & \text{otherwise} \end{cases}$$

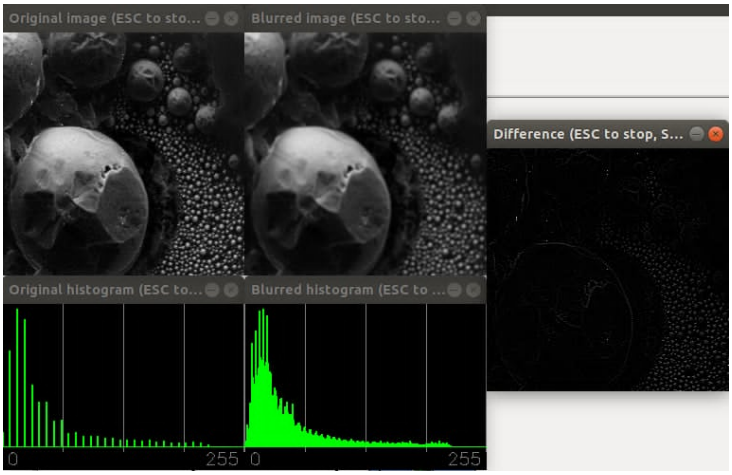
The resulting image is binary. Therefore the histogram contains only two bars at intensities 0 and 255, reflecting the two pixel classes produced by the threshold.

3 Part 3 - Linear and non-linear filters

3.1 Mean (moving average) filter

Exercise (a)

1 - Display meb.bmp image histogram. Can you deduce the number of bits used to digitize this image ?



There are approximately 29 levels displayed in the histogram while the number of intensity levels $= 2^n$ where n is the number of bits used per pixel. $2^5 = 32$ closest to 28–29. So the image was likely digitized using 5 bits per pixel.

2 - Apply a 3x3 mean filter (blur) to this image. Display the resulting image histogram. Explain its shape.

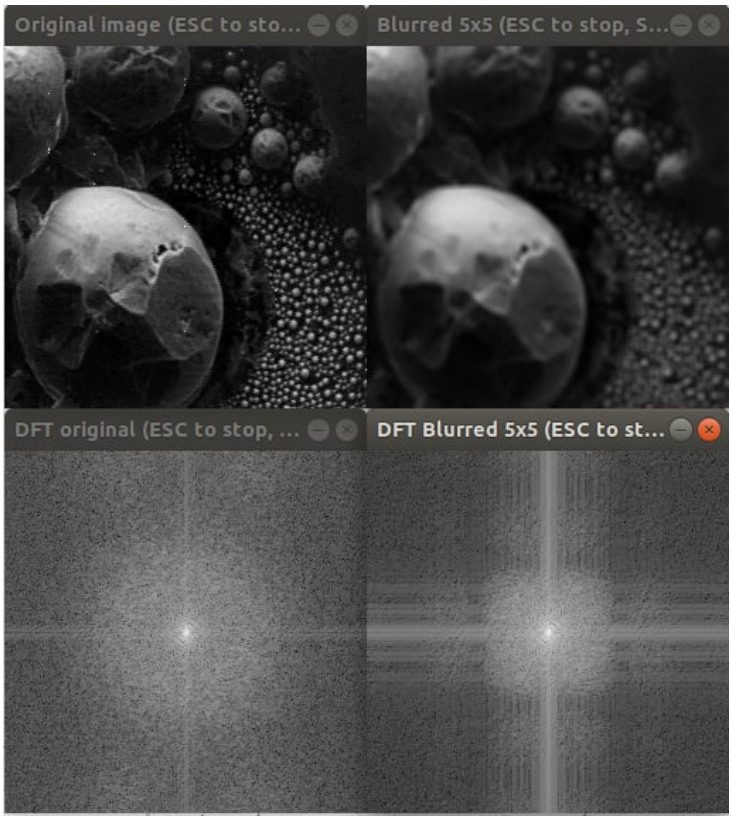
Each pixel is replaced by the average of itself and its 8 neighbors ($= 3 \times 3$), making the image blured:

- The histogram becomes more complete and wider, having extra intermediate intensity values
- Peaks of the original histogram are reduced, valleys are filled. Overall more continuous distribution

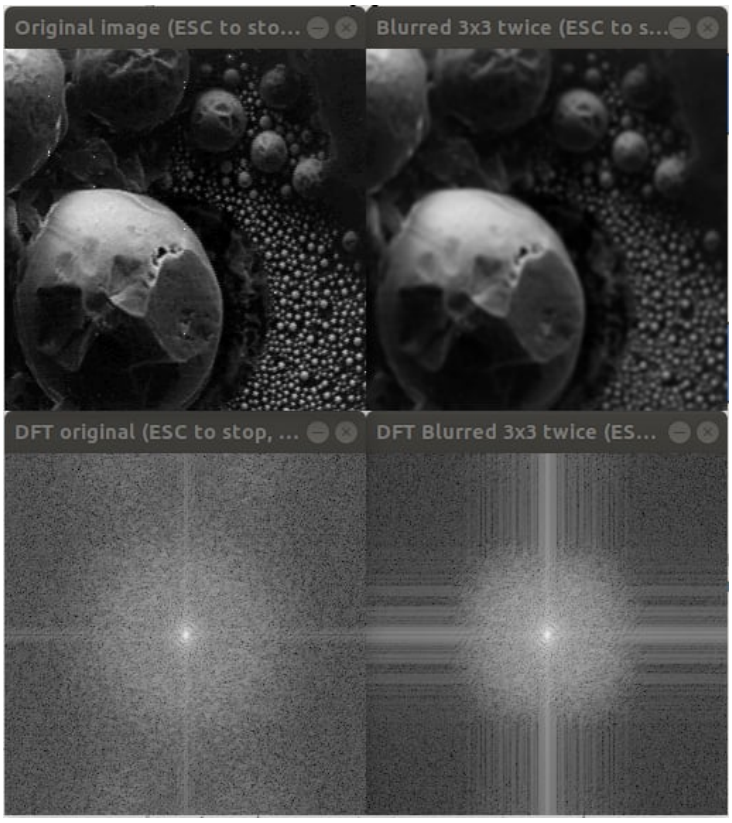
3 - Compute the difference between the original and the blurred image. Explain the result. The difference displayed are the extra 8 neighbors of the pixels computed. However, only those of the brighter shades are visible to human eyes, indicating that the filter replace and add extra intenisity levels to smooth intensity variations.

Exercise (b)

1 - Apply a 5x5 mean filter to image meb.bmp.

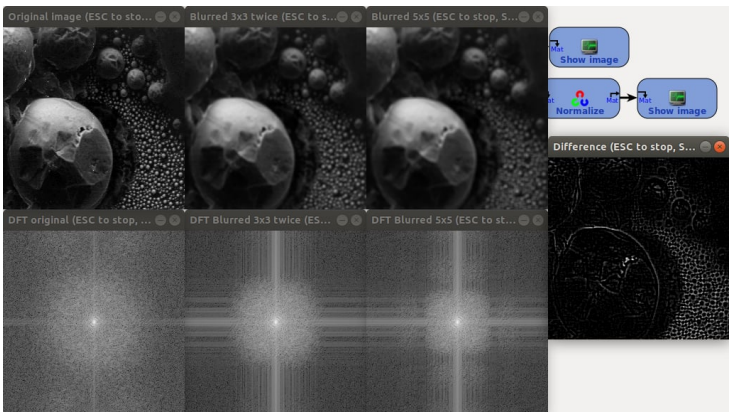


2 - Apply two consecutive 3x3 mean filters to image meb.bmp.



3 - Are resulting images similar ? Why ? Which one of the two operations removes high frequencies more efficiently ? Compute the Discrete Fourier Transform on the

original image and the two filtered images, and explain the result.



Spatial-domain comparison of the two filters

A 3×3 mean filter has kernel:

$$h_{3\times 3} = \frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Applying this filter twice means convolving the kernel with itself. In 1D, the convolution of $[1, 1, 1]$ with $[1, 1, 1]$ gives:

$$[1, 1, 1] * [1, 1, 1] = [1, 2, 3, 2, 1]$$

After normalization:

$$h_{1D} = \frac{1}{9}[1, 2, 3, 2, 1]$$

This produces a non-uniform 5×5 kernel where central pixels have larger weights and border pixels have smaller weights. This structure is closer to a Gaussian blur than to a mean blur.

By contrast, a real 5×5 mean filter is:

$$h_{5\times 5} = \frac{1}{25}\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix}$$

This kernel gives equal weight to all 25 pixels, unlike the effective kernel produced by two 3×3 passes.

Which filter removes high frequencies more efficiently?

- The 5×5 mean filter removes a larger band of high frequencies because it averages over a larger neighbourhood with uniform weights.
- The two 3×3 filters also remove high frequencies, but because their effective kernel is Gaussian-like, they attenuate very high frequencies strongly but attenuate mid-range frequencies slightly less than the uniform 5×5 kernel.

$\Rightarrow$ In a single operation, the 5×5 mean filter suppresses high frequencies more uniformly and more strongly.

Discrete Fourier Transform (DFT) analysis

Let $F(u, v)$ be the DFT of the original image.

- **Original image:** Significant energy appears in both low and high frequencies (edges, fine textures).
- **After 5×5 mean filter:** The DFT spectrum shows strong attenuation in the high-frequency region (outer ring). Low frequencies dominate, indicating effective smoothing.
- **After two 3×3 mean filters:** High frequencies are attenuated, but the decay is smoother. Compared to the 5×5 mean:
  - very high frequencies are similarly suppressed,
  - mid frequencies retain slightly more energy (because the kernel is not uniform).

Conclusion

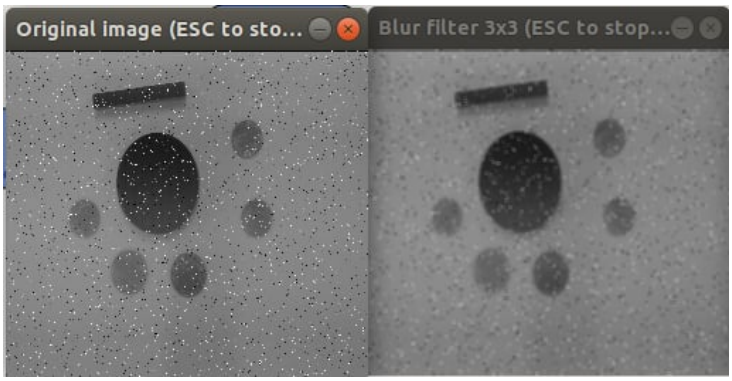
The two operations do not produce identical filtering:

- A true 5×5 mean performs stronger, uniform high-frequency attenuation.
- Two 3×3 means produce a Gaussian-like kernel with a smoother transition in frequency space.

This explains the differences observed in the spatial and frequency domains.

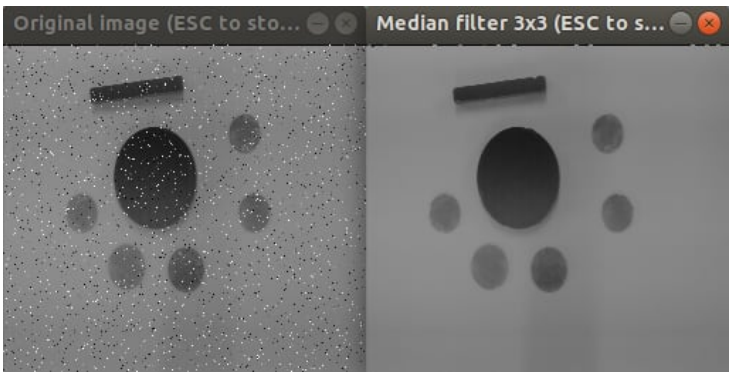
3.2 Non linear filter

1 - Apply a 3x3 mean filter to morpho_salt5.bmp image, which has been degraded by a 5% pepper and salt noise



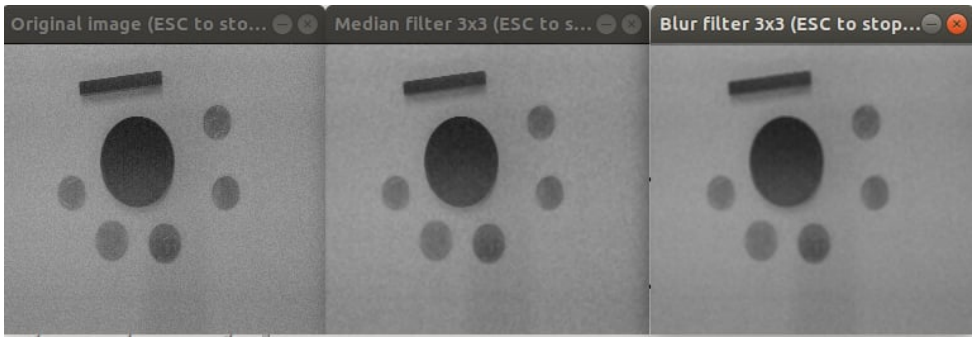
The 3×3 mean filter slightly reduces the salt-and-pepper noise, but many black and white specks remain visible. This is because the mean is strongly influenced by the extreme values 0 and 255, so noisy pixels still affect the average and are not completely removed.

2 - Apply a 3x3 median filter to morpho_salt5.bmp image. Compare the results of the two filters.



The 3×3 median filter removes most of the salt-and-pepper noise while preserving the shapes of the objects. Unlike the mean, the median is robust to outliers: isolated 0 or 255 pixels do not influence the median value of the neighbourhood. As a result, noisy pixels are replaced by values consistent with their neighbours, and edges remain relatively sharp.

3 - Do the same operations with morpho_unif20.bmp image, which has been degraded with an uniform noise of magnitude 20. What do you conclude ?

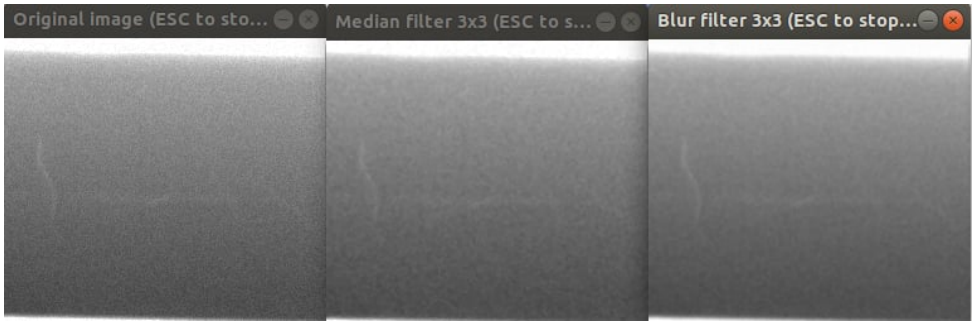


For the uniformly noisy image, the mean filter produces a smoother result than the median filter. Uniform noise slightly perturbs almost all pixels, and there are no isolated extreme values. In this situation, averaging neighbouring pixels reduces the random fluctuations efficiently, whereas the median filter produces only a small change because neighbouring values are already similar.

4 - Repeat steps 1 to 3 with images fissure_salt5.bmp and figure_unif20.bmp. In which cases should we use each filter ?

Figure_salt5.bmp

Figure_unif20.bmp



Median filter - Salt-and-pepper noise

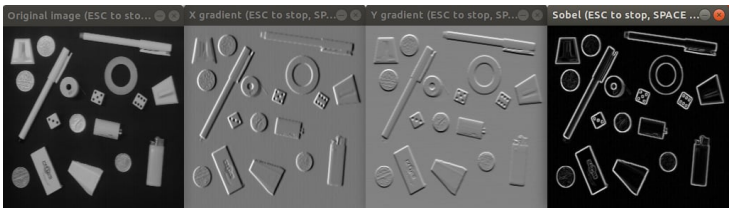
The median filter is best for salt-and-pepper noise because it removes isolated outliers while preserving edges.

Mean filter - Uniform noise

The mean filter is best for uniform noise because it averages small fluctuations.

3.3 Edge detection linear filters

1 - Compute X gradient, Y gradient and gradient norm on image objects.bmp, using the Sobel operator



2 - Comment the result The Sobel X filter highlights vertical edges (intensity changes along the horizontal direction), while the Sobel Y filter highlights horizontal edges. The gradient norm combines both components and gives the strongest response at the contours of the objects, producing a clear edge map.

3.4 Canny operator

Canny edge detector

The Canny edge detector is a multi-stage edge detection algorithm. It is designed to detect meaningful edges in images while minimizing the influence of noise and avoiding multiple responses

for a single edge.

The method was designed to satisfy three main criteria:

- Good detection: all real edges should be detected with a low probability of false detection.
- Good localization: detected edges should be as close as possible to the true edges.
- Single response: each edge should produce only one response.

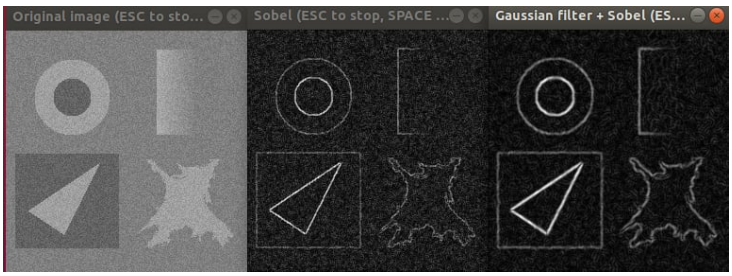
To achieve these objectives, the Canny operator follows these processing steps:

1. **Noise reduction**
   Since edge detection is sensitive to noise, the image is first smoothed using a Gaussian filter. This step reduces high-frequency noise that could otherwise generate false edges.
2. **Gradient computation**
   After smoothing, the intensity gradients in the horizontal and vertical directions are computed, usually using Sobel operators. From these gradients, both the gradient magnitude and gradient direction are obtained, indicating the strength and orientation of edges.
3. **Non-maximum suppression**
   To obtain thin edges, the algorithm keeps only the pixels that are local maxima of the gradient magnitude in the direction of the gradient. All other pixels are suppressed, resulting in nearly one-pixel-wide edges.
4. **Double thresholding**
   Two thresholds are applied: a high threshold and a low threshold. Pixels with gradient magnitude above the high threshold are classified as strong edges. Pixels with gradient magnitude between the two thresholds are classified as weak edges, while pixels below the low threshold are discarded.
5. **Edge tracking by hysteresis**
   Weak edge pixels are preserved only if they are connected to strong edge pixels. This step eliminates isolated responses caused by noise while keeping continuous edge contours.

Exercise (a)

1 - Apply the Sobel operator to bruit.bmp image.

2 - Apply the Sobel operator to bruit.bmp image filtered with a Gaussian filter.



3 - On which image are the edges more clearly visible ? When Sobel is applied directly to bruit.bmp, the result is noisy and many spurious edges appear. After applying a Gaussian smoothing before Sobel, the noise is reduced and the edges become cleaner and more continuous. This illustrates that gradient-based edge detectors work better when high-frequency noise is removed first.

The edges are clearer with the compbination of the Sobel operator and the Gaussian filter. Applying a Gaussian filter before Sobel reduces noise, which results in cleaner and more continuous edges in the gradient magnitude image.

Exercise (b)

1 - Apply the Sobel operator to bruit.bmp image.

2 - Apply the Canny operator to bruit.bmp image.

3 - Change the Canny thresholds (one at a time) and explain what happens. Canny uses two thresholds:

- T_{low} (weak edge threshold):
  - Increase T_{low}: missing small/weak edges, clean but incomplete
  - Decrease T_{low}: more edges appear, including edges which are not important
- T_{high} (strong edge threshold):
  - Increase T_{high}: fewer edges, only the strongest contours remain
  - Decrease T_{high}: more edges appear, including edges which are not important

Canny edge detector method summary

1. **Gaussian smoothing**
   The image is first convolved with a Gaussian kernel

$$G(i,j) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{i^2+j^2}{2\sigma^2}\right)$$

   to reduce noise.

2. **Gradient magnitude and direction**
   Sobel filters are used to compute

$$G_x,\ G_y,\ |G| = \sqrt{G_x^2 + G_y^2},\ \theta = \arctan\left(\frac{G_y}{G_x}\right).$$

3. **Non-maximum suppression**
   For each pixel, the gradient magnitude is compared with its neighbours along the gradient direction. Only local maxima are kept; other pixels are set to zero. This produces thin edges.

4. **Double thresholding**
   Two thresholds are used:

$$|G| \geq T_{\text{high}} \Rightarrow \text{strong edge}$$

$$T_{\text{low}} \leq |G| < T_{\text{high}} \Rightarrow \text{weak edge}$$

$$|G| < T_{\text{low}} \Rightarrow \text{non-edge}$$

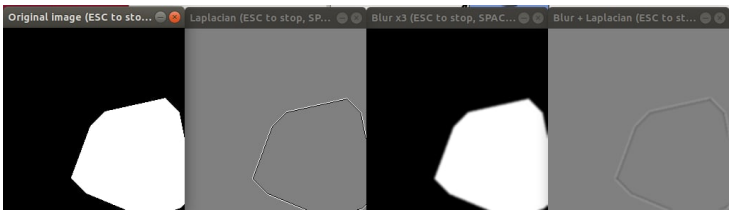
5. **Edge Tracking by Hysteresis**
   Weak edges (pixels with gradient magnitude between the low and high thresholds) are preserved only if they are connected to strong edges. Any weak edges not connected to strong edges are suppressed, ensuring that only meaningful edges remain in the final output.

3.5 Laplace operator

Exercise (a)

1 - Apply the Laplace operator on spot.bmp image.

2 - Apply the mean filter three times on spot.bmp image, then apply the Laplace operator. Comment the result.



Before the mean filter three times

Without pre-smoothing, the Laplacian responds very strongly to the sharp transition between the spot and the background, producing very bright, thin contours but also amplifying any noise.

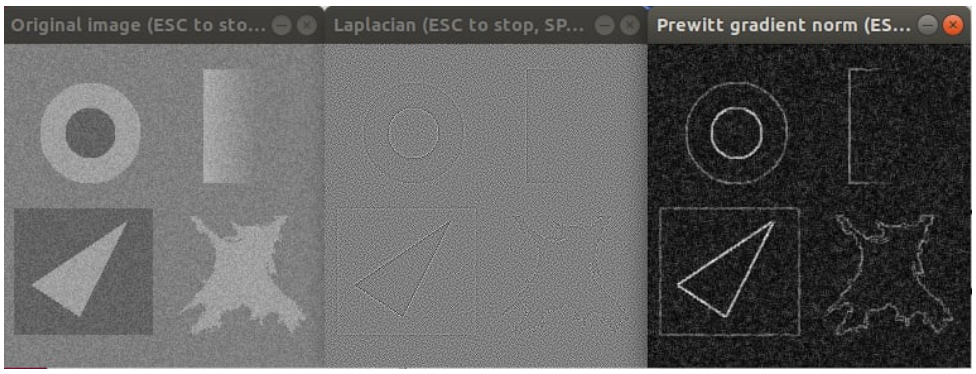
With the mean filter three times

After applying the mean filter three times, the image becomes smoother and the intensity transition at the edges is less abrupt. The Laplacian response is therefore weaker and the edges appear thicker and less sharp. Pre-smoothing reduces noise but also reduces the Laplacian edge strength.

Exercise (b)

1 - Apply the Laplace operator on bruit.bmp image

2 - Compute the gradient norm (using Prewitt filter) on bruit.bmp image. Which one of filtered images can you detect edges on more easily ? Why ?



- The Laplace image contains many strong responses due to noise because the Laplacian is a second-order derivative and is very sensitive to high-frequency components.
- The Prewitt gradient norm, based on first-order derivatives, is less sensitive to noise and mainly highlights meaningful transitions. The resulting edge map is therefore easier to interpret.

Exercise (c)

1 - Apply the Laplace operator on film.bmp image.

2 - Combine the result with 60% of the original image. Explain the result.

The Laplacian image contains mainly high-frequency information (edges). When adding a scaled version of this Laplacian to the original image (60%), the edges are reinforced while the low-frequency content is preserved. This operation is a form of Laplacian-based sharpening: contours become more pronounced but the overall brightness of the image remains similar.

4 Part 4 - Mathematical morphology

Exercise (a)

1 - Erode and dilate `morpho_math.bmp` separately using a 5x5 rectangular structuring element.

2 - For better visualization, resize erosion and dilation outputs to 4 times the original image size before displaying them. Display the structuring element as well.

Figure 8: Erosion, dilation and structuring element

3 - What do the operations erode and dilate do? Try to give a formal definition.

Erosion:

- Shrinks bright regions (foreground) in a binary or grayscale image
- Removes pixels from object boundaries
- Small objects may disappear completely
- Objects get thinner
- Noise which is smaller than the structuring element is removed

Dilation:

- Expands bright regions (foreground) in a binary or grayscale image
- Adds pixels to object boundaries
- Small holes or gaps inside objects are filled
- Objects grow in size
- Noise may also expand if present

3 - What do the operations erode and dilate do? Try to give a formal definition (set-based).

Let A be a binary image and B a structuring element.

Erosion

The erosion of A by B is defined as:

$$A \ominus B = \{x \mid B_x \subseteq A\}$$

where B_x is the structuring element translated by x.

A pixel in the output remains white only if the entire structuring element fits inside the object. Erosion:

- shrinks objects,
- removes small white regions,
- separates thin connections.

Dilation

The dilation of A by B is defined as:

$$A \oplus B = \{x \mid (B^{\text{ref}})_x \cap A \neq \emptyset\}$$

where B^{ref} is the reflected structuring element.

A pixel becomes white if any part of the structuring element overlaps with a white pixel of the image. Dilation:

- expands objects,
- fills small holes,
- thickens boundaries.

4 - Experiment with different structuring element types and sizes, and different numbers of erosion and dilation iterations. Note down and explain your observations

Figure 9: Examples of erosion and dilation with different parameters (1)

Figure 10: Examples of erosion and dilation with different parameters (2)

Structuring element	Size	Iterations	Effects
Rectangular	3x3	1	**Erosion:** Slight shrinkage of object boundaries. **Dilation:** Slight expansion; preserves corners.
Rectangular	5x5	1	**Erosion:** Noticeable shrinkage. **Dilation:** Noticeable expansion; gaps partially fill.
Rectangular	5x5	3	**Erosion:** Strong shrinkage; small objects vanish. **Dilation:** Strong expansion; small gaps fill; effects accumulate with iterations.
Elliptical / Circular	5x5	1	**Erosion:** Smoother shrinkage; corners round. **Dilation:** Smooth expansion; rounded edges, less abrupt changes.
Cross-shaped	5x5	1	**Erosion:** Shrinks mainly along horizontal and vertical directions. **Dilation:** Expands mainly along horizontal and vertical; diagonal features less affected.
Any structuring element	7x7, 9x9	Multiple	**Erosion:** Very strong shrinkage; small details disappear. **Dilation:** Very strong expansion; small holes fill completely due to size + iterations.

Exercise (b)

1 - Start from the processing chain of the previous exercise using the same image and the structuring element as inputs.

2 - Erode the negative of the input image and display it. What do you conclude?

Erosion on a binary image shrinks white regions. If we take the negative of the input, white and black regions are swapped.

Therefore:

$$(A^c \ominus B) = (A \oplus B)^c$$

Figure 11: Erosion of the negative image

Eroding the negative of an image is equivalent to taking the complement of the dilation of the original image.

This explains why the result looks like the inverse effect of dilation: small black regions shrink, while white regions expand visually after inversion.

Exercise (c)

1 - Start from the processing chain of Exercise (a) using the same image and the structuring element as inputs.

2 - Dilate the output of the previous erosion, and erode the output of the previous dilation. Visualize outputs.

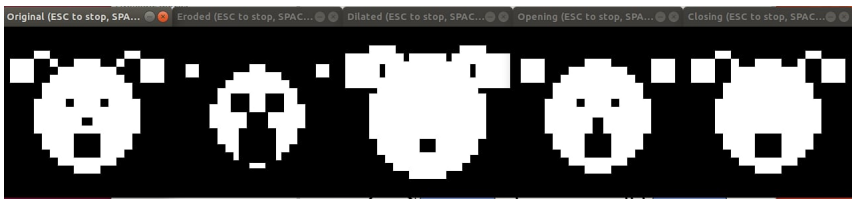


Figure 12: Opening and closing results

3 - What are the combined operations called? Try to define them intuitively and explain the effect of each

Operation	Intuitive Definition / Effect
Opening (erosion to dilation)	Removes small bright noise while preserving the overall shape of larger objects
Closing (dilation to erosion)	Fills small dark holes or gaps, connects nearby bright regions, while keeping object edges intact

Opening Opening is defined as:

$$A \circ B = (A \ominus B) \oplus B$$

It first erodes the image (removing small white regions), then dilates it (restoring the main structure).

Effect:

- removes small bright objects,
- smooths contour irregularities,
- separates thin connections,
- preserves the overall shape of large objects.

Closing Closing is defined as:

$$A \bullet B = (A \oplus B) \ominus B$$

It first dilates the image (filling holes), then erodes it (restoring original object size). Effect:

- fills small dark holes,
- connects nearby objects,
- smooths dark gaps and cracks.

5 Part 5 - Image segmentation

5.1 Segmentation based on automatic thresholding

1 - Apply the automatic segmentation on image rondelle.bmp, then comment the result

Otsu's method automatically chooses the threshold that maximizes the between-class variance of the histogram. For *rondelle.bmp*, the histogram is clearly bimodal: one peak corresponds to the dark background and the other corresponds to the brighter ring-shaped object.

The threshold selected by Otsu lies near the valley between these peaks. As a result, the segmentation cleanly isolates the ring from the background, producing a binary image where the object is clearly separated from its surroundings.

2 - Do the same with image circuit.bmp

The segmentation extracts the main circuit elements from the background. Edges are less clear and some fine details are partially lost. Works well but only on parts where the contrast are distinctive.

The histogram of *circuit.bmp* is not cleanly bimodal. The wires, the circular nodes, and the background have overlapping gray-level distributions. Otsu's method chooses a single global threshold, which is not ideal for an image containing several materials with similar intensities.

The segmentation therefore detects most circular nodes, but many wires are only partially segmented or lost entirely. Some background areas are incorrectly classified as foreground. This illustrates the limitation of global thresholding for images with non-uniform illumination or overlapping intensity distributions.

5.2 Split and merge segmentation

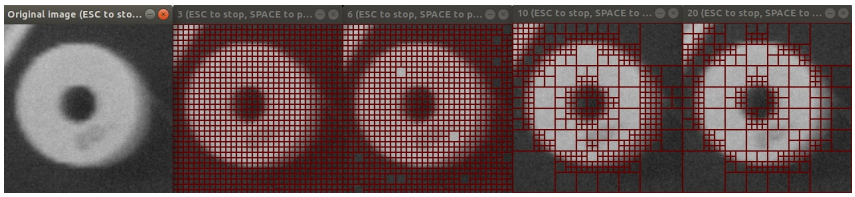
Exercise (a)

1 - Apply the split segmentation step on image des.bmp

2 - Explain the resulting image

The split algorithm recursively divides the image into smaller square regions. A region is split if it does not satisfy a homogeneity criterion, typically based on intensity variance. In *des.bmp*, regions containing points or edges have high variance and are therefore subdivided, while uniform background regions remain un-split. The resulting image shows many small blocks around the dices and larger blocks in uniform areas.

3 - Apply the split segmentation on rondelle.bmp. Try the following values of the split parameter: 3, 6, 10, 20.



Explain the results. What is the more appropriate value of the parameter ?

The split parameter controls the homogeneity threshold.

- **Threshold = 3:**
  Very strict condition. Almost every region is considered non-homogeneous, so the image is over-segmented into many tiny blocks.

- **Threshold = 6:**
  Fewer blocks; smooth areas begin to merge while edges still cause subdivision.

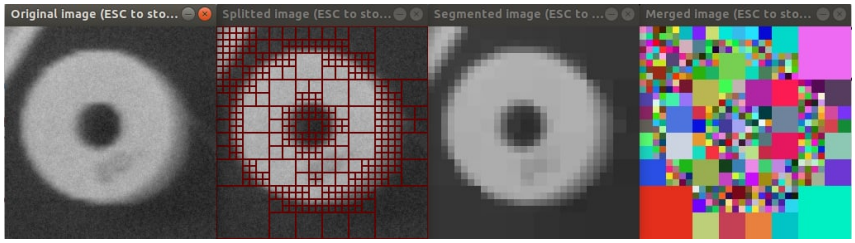
- **Threshold = 10:**
  Good compromise. The ring structure is represented with appropriate subdivisions while uniform background remains in large regions.

- **Threshold = 20:**
  The criterion becomes too tolerant: even regions containing edges are considered homogeneous. The segmentation becomes almost meaningless.

Best value: Between 10 and 20, because it captures the object structure without over-segmenting the background.

Exercise (b)

1 - Apply split and merge segmentations on rondelle.bmp.



2 - What do you think of the result of the merge step compared to the split step ?

The split step divides the image into many small regions, while the merge step combines adjacent regions with similar intensity.

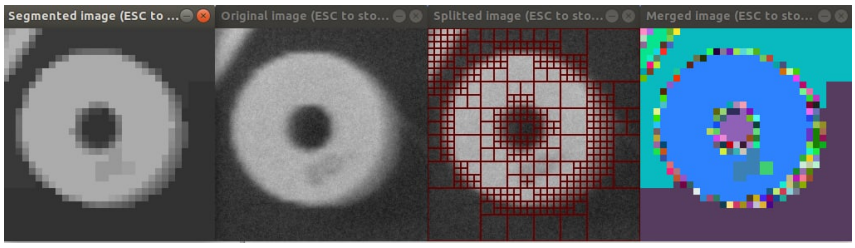
How can you explain this result ?

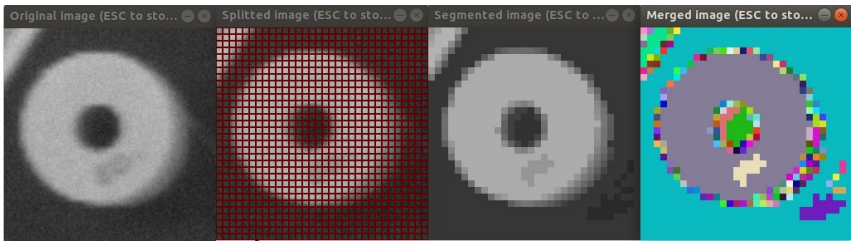
- The split step divides the image into many small regions. The merge step is intended to combine adjacent regions with similar intensity.

- In this case, the merge parameter is too low, so the regions of merged image are almost identical to those of the split image.

- The regions remain as small and fragmented as after splitting, and over-segmentation is not effectively reduced.

3 - Find a better value for the merge parameter and explain the result.





In this case, using a merge parameter of 4 is better than 10.

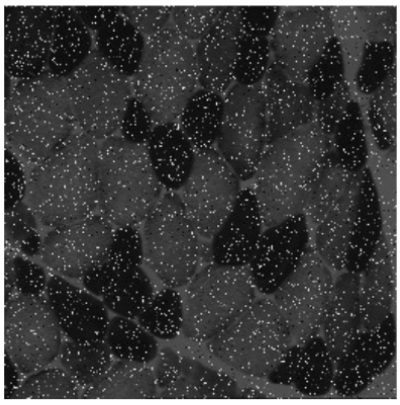
- With 10, the merge is too aggressive, and distinct regions that should remain separate are combined, losing detail.
- With 4, regions are smaller and better preserve the image structure, making the segmentation more accurate while still reducing over-segmentation from the split step.

4 - How do we obtain the final segmented image ? What is the meaning of the different gray level values ?

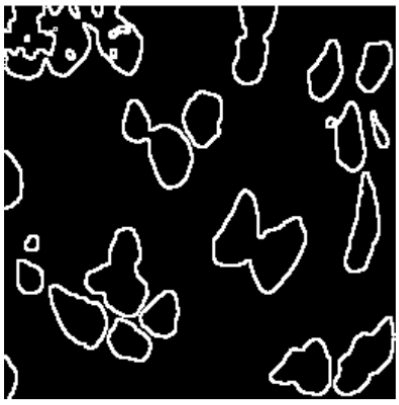
- The final segmented image is obtained after the split and merge steps.
- Each region is assigned a different gray level to distinguish it from other regions.
- The gray level values do not represent intensity but label different segments: pixels with the same gray value belong to the same region.

6 Final exercise 1

The original image `MUSCLE_BAD_QUALITY.BMP` is strongly degraded by salt-and-pepper noise and low contrast. The goal of this exercise is to detect and extract the contours of the dark gray fibres, as in the target image provided in the lab instructions.

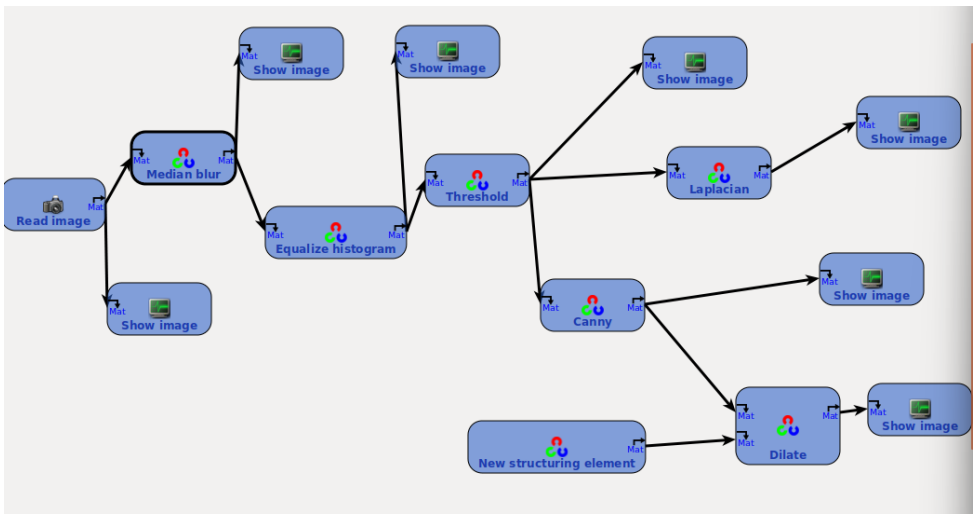


Original image



Result to be obtained

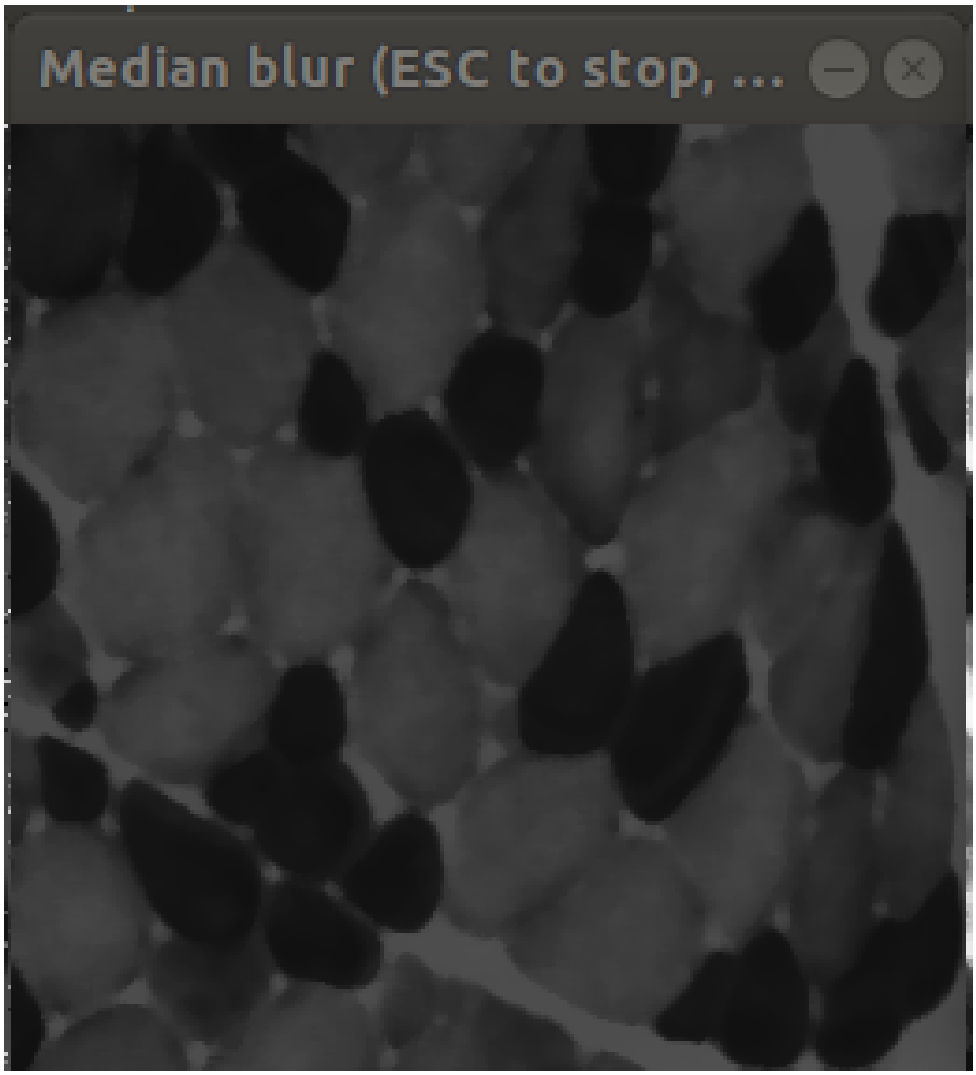
6.1 The processing chain



The processing chain is summarized in the diagram above. It is composed of the following blocks:

- Noise reduction using a median blur.
- Histogram equalization to enhance contrast.
- Fixed thresholding to isolate dark fibres.
- Two alternative edge detectors applied on the thresholded image: Laplacian and Canny.
- Morphological dilation of the Canny edges to obtain contours similar to the expected result.

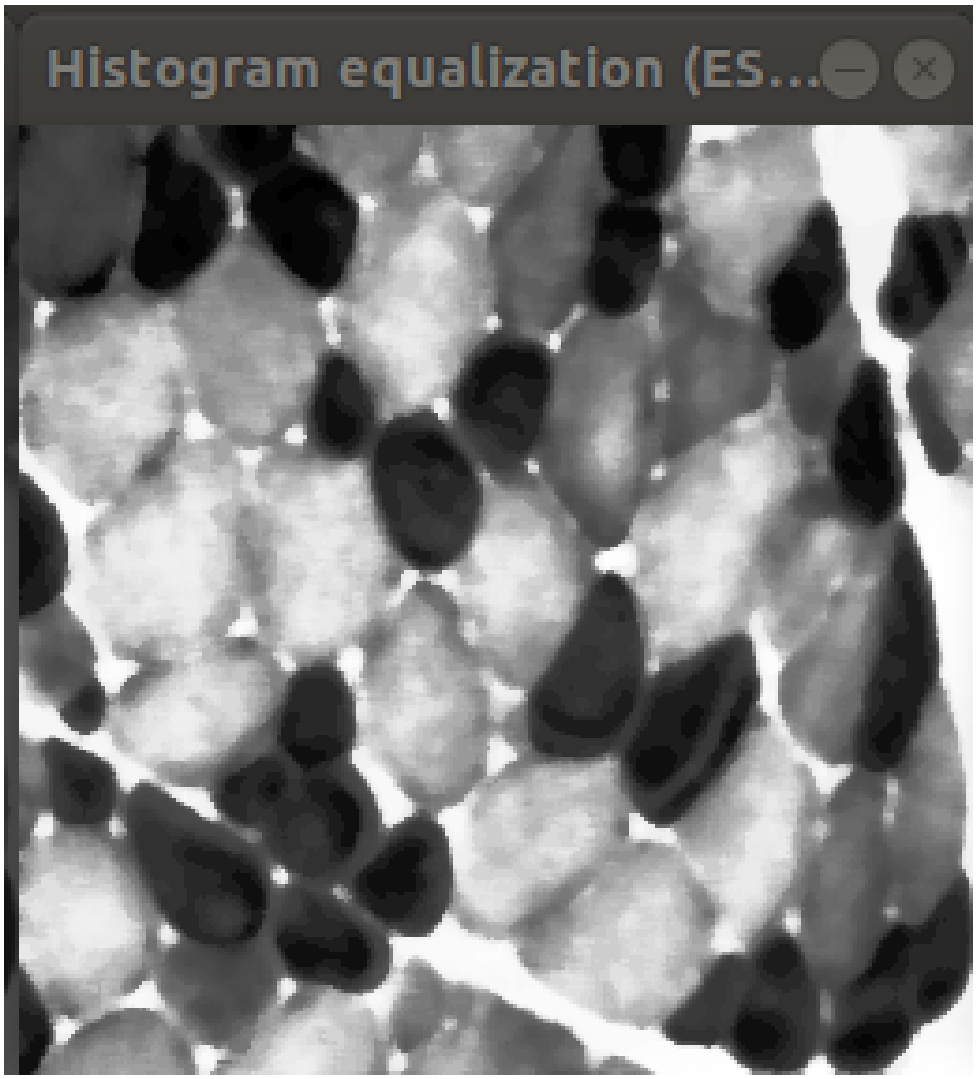
6.2 Step 1 : Noise reduction (median blur)



The original image is heavily corrupted by white specks due to salt-and-pepper noise. To remove this noise while preserving the shapes of the fibres, We apply a median blur with an aperture size of 5 (5×5 window).

The median filter is particularly well suited for salt-and-pepper noise: in each neighbourhood it replaces the central pixel by the median value, which is robust to isolated outliers. As a result, most of the small bright dots disappear, whereas the boundaries of the dark fibres remain sharp.

6.3 Step 2 : Contrast enhancement (histogram equalization)

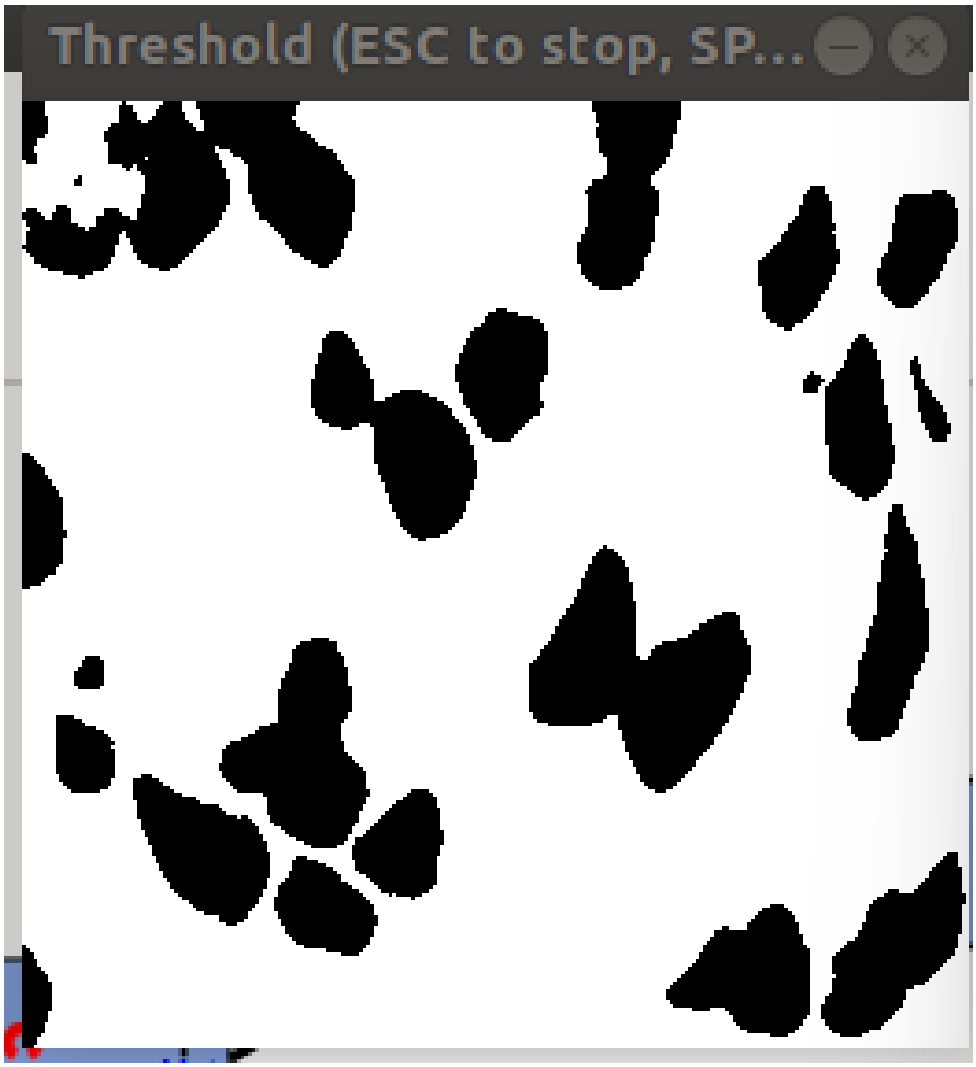


After noise reduction, the contrast between the dark fibres and the background is still relatively low. We then apply histogram equalization to the blurred image.

Histogram equalization redistributes the gray levels so that the output histogram is approximately uniform. This stretches the dynamic range and increases the contrast: dark fibres become darker, bright background becomes brighter, and the transitions between them are more pronounced.

This step makes it easier to choose a threshold that separates fibres from the background.

6.4 Step 3 : Thresholding



From the histogram of the equalized image, we select a threshold value $T = 64$ with type `THRESH_BINARY`. Pixels with intensity lower than 64 (dark fibres) are mapped to 0. Here the parameters are chosen so that the fibres appear as white regions on a black background.

The thresholded image is a binary mask where the interior of the dark fibres forms large homogeneous regions. This simplifies the subsequent edge detection, because edges now correspond to transitions between 0 and 255 instead of subtle gray-level changes.

6.5 Step 4a : Laplacian

As a first approach, we compute the Laplacian of the thresholded image. The parameters are:

- destination depth: -1 (same type as input image)
- aperture size: 1
- scale: 1, delta: 0

The Laplacian is a second-order derivative operator. On a binary image, it responds strongly at positions where the intensity changes abruptly from 0 to 255, i.e. at region boundaries. The result already shows thin contours around the fibres, but the operator is quite sensitive to small irregularities and tends to produce very thin, sometimes broken lines. This motivated the use of a second method, Canny, which includes smoothing and non-maximum suppression.

6.6 Step 4b : Canny edge detector + Dilation

We then apply the Canny edge detector to the thresholded image with the following parameters:

- first threshold (low): 90
- second threshold (high): 150
- Sobel aperture size: 3

Canny performs Gaussian smoothing, gradient computation, non-maximum suppression and hysteresis thresholding. Compared to the Laplacian, it produces cleaner and more continuous contours, with less sensitivity to small noise and local fluctuations. The resulting edges already resemble the target, but they are slightly thinner and may contain small gaps.

To obtain contours that visually match the expected result, we slightly thicken the Canny edges using a morphological dilation. We use a small structuring element (3×3) and one iteration.

Dilation adds pixels around each edge point inside the structuring element support. This has two effects:

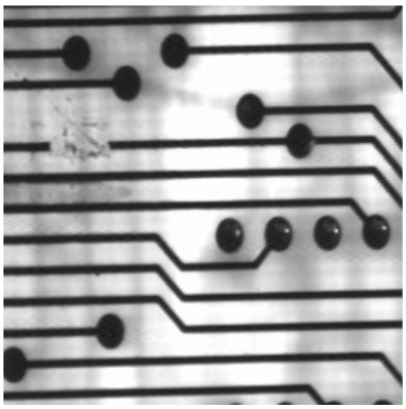
- edges become thicker and more visible
- small gaps in the contours are closed, leading to more continuous boundaries around each fibre.

The final image shows bright closed contours around the dark fibres on a black background, which corresponds closely to the reference result provided in the statement.

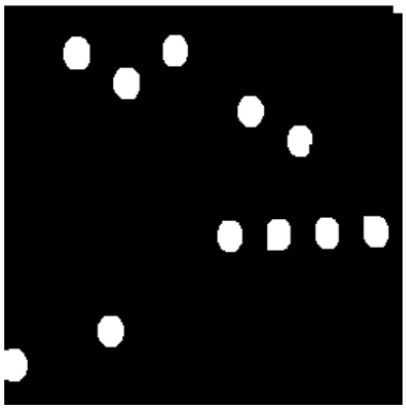
Although both Canny and Laplacian can be used for edge detection, Canny is more appropriate for this exercise. The Laplacian operator is very sensitive to noise and produces thin, unstable contours in general cases. Canny integrates Gaussian smoothing, non-maximum suppression and hysteresis, which allows it to extract clean, continuous and well-defined contours of the dark fibres. Therefore, Canny was the most reliable choice for obtaining the expected result.

7 Final exercise 2

The goal of this exercise is to clean printed-circuit image using morphological operations and to extract only the circular solder pads while removing wires.

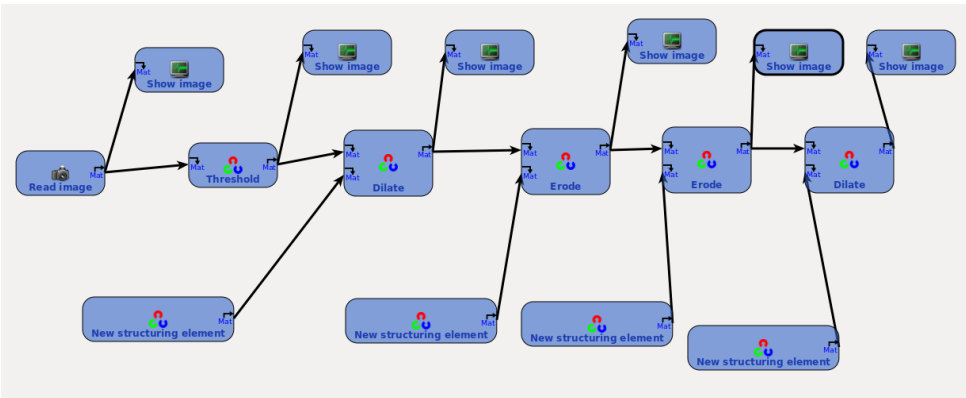


Original image



Result to be obtained

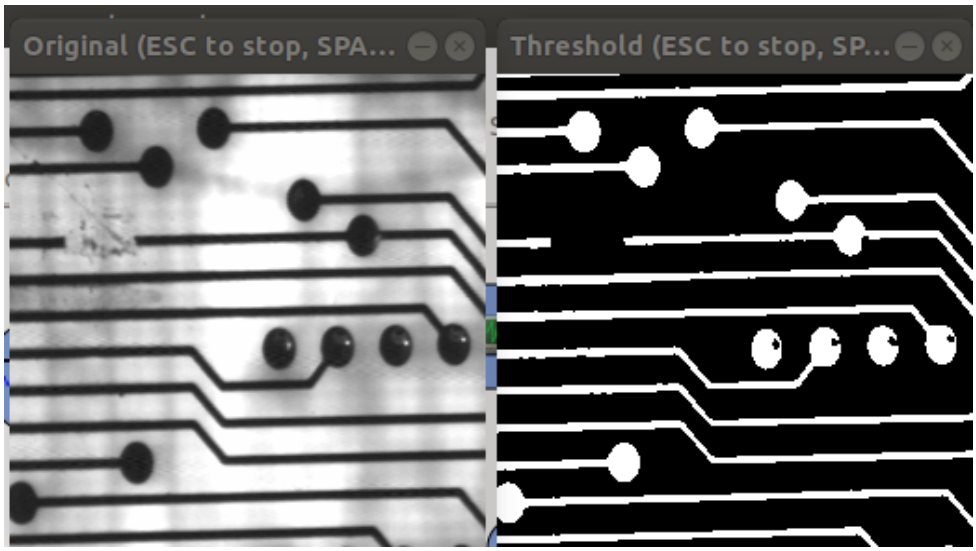
7.1 Processing chain



It is composed of the following blocks:

- Fixed thresholding to isolate all bright regions (wires + circular nodes).
- Closing (Dilation → Erosion) to merge and clean the wires and reduce small discontinuities.
- Opening (Erosion → Dilation) to remove the wires while keeping the circular nodes intact.

7.2 Thresholding the original image



We apply a fixed threshold of 70 using `THRESH_BINARY_INV`.

In this image, the circular nodes are bright, and the background/wires vary in brightness. Using an inverted threshold turns bright circular nodes into white blobs, while wires and background become black.

The result is a binary image containing all round features, but wires still appear as thin white lines.

By doing that the thresholding simplifies the segmentation by converting the grayscale image into a binary mask containing only potential objects of interest in our case the circular nodes.

7.3 Closing (dilation → erosion)

Structuring element used:

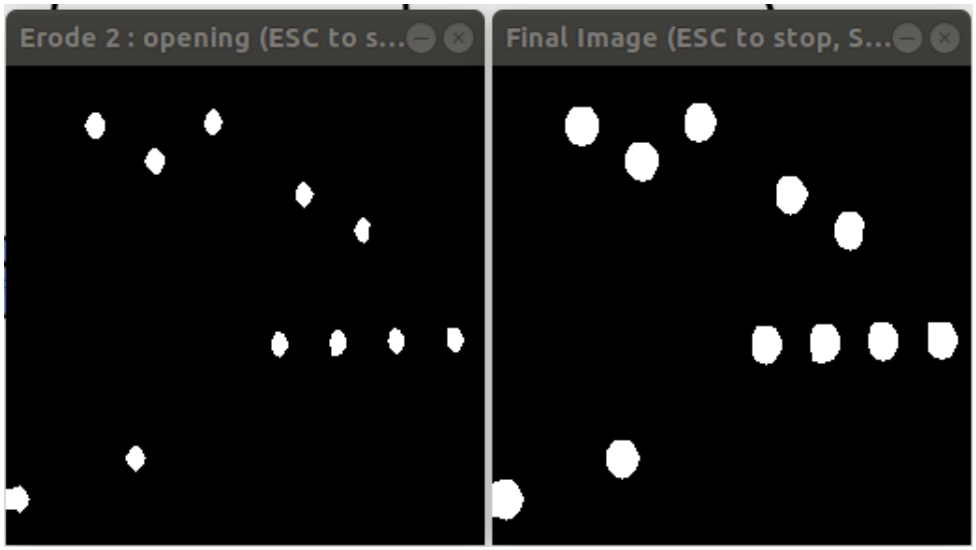
- Dilation: RECT, size (4,4)
- Erosion: RECT, size (4,4)

Doing that caused the fills of small gaps and connects discontinuities in the thresholded shapes. Here, it:

- thickens the thin wire segments
- merges nearby pixels inside wire regions,
- ensures that wires become continuous white regions.

This prepares the image for the next step, removing these unwanted linear structures through opening.

7.4 Opening (erosion → dilation)



Structuring elements used:

- Erode: ELLIPSE (8,8)
- Dilate: ELLIPSE (8,8)

Opening removes objects that are thin or narrow compared to the structuring element.
What happens?

- Wires are long thin structures, after erosion with a large enough kernel, they disappear completely.
- Circular nodes, because they are round and significantly larger, survive the erosion.

The final dilation step restores their original size and shape.
Why elliptical SE?
An ellipse is isotropic, which preserves circular shapes better than a rectangular kernel, avoiding geometric distortion of the nodes (to keep them as round as possible).

References

[1] Canny edge detector, Wikipedia: https://en.wikipedia.org/wiki/Canny_edge_detector, accessed 12 Dec 2025.